Assignment 11 Performance Analysis of a Heterogeneous Wireless Network

Deadline: 05/10/26 & 08/10/26

Consider a smart-campus network consists of **Wi-Fi, Bluetooth, and Zigbee nodes** communicating with their respective access/coordinator nodes. Mobile users move across the campus, causing changes in connectivity and signal quality. Design this scenario using simulation in NS3.

a. Network Setup

- Create Wi-Fi, Bluetooth, and Zigbee nodes.
- Configure appropriate wireless channels, PHY/MAC protocols, IP addressing, and applications.
- Establish communication between mobile nodes and their corresponding gateways.

b. Mobility

- Assign mobility models such as **RandomWalk** or **RandomWaypoint** to selected nodes.
- Vary the node speed, for example, **1 m/s, 5 m/s, and 10 m/s**.

c. Performance Measurement

Measure the following for each mobility condition:

1. **Signal Strength (Received Power)** – strength of the received wireless signal.
2. **Throughput** – amount of successfully delivered data per second.
3. **Packet Loss** – percentage of packets that fail to reach the destination.
4. **End-to-End Delay** – time taken for a packet to reach the destination.

d. Analysis

Compare the results for different node speeds and **analyze how increasing mobility affects signal strength, throughput, packet loss, and delay** for the different wireless technologies.